# PROJECT PROPOSAL: SMART DOCUMENT SCANNER & CHATBOT ASSISTANT

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1. **Objective**

To develop a web-based platform that enables users to:

Upload important documents (receipts, invoices, certificates)

Automatically extract and analyze text content using OCR

Interact with a chatbot that can understand and respond based on document content

Export extracted data in structured formats (CSV, JSON)

Maintain a history of document uploads and queries

Ensure mobile responsiveness and ease of use

2. **Key Features**

1. Document Upload Interface

2. OCR Processing (Tesseract )

3. Data Extraction and Structuring

4. Al Chatbot to Query Document Content

5. Export Results (CSV / JSON)

6. User Authentication and Management

7. Mobile-Responsive Web Design

3. **Technologies Used**

Component Technology

Frontend React.js + TailwindCSS
Backend FastAPI (Python)
OCR Engine Tesseract OCR / PaddleOCR
Chatbot LangChain + HuggingFace Model (Mistral / Phi-2)
Storage Firebase Storage or MongoDB
Authentication Firebase Auth
Hosting Vercel (Frontend) + Render (Backend)

4. **System Architecture**

- User Uploads Document
- React.js Frontend
- FastAPI Backend
    - Tesseract OCR
    - Text Parsing & Structuring
    - LangChain + LLM Chatbot
- Response + Download + Export

5. Implementation Timeline (20 Days)

Phase Days Tasks

1 Day 1-3 Project setup, basic UI, Firebase
Auth 2 Day 4-6 OCR integration (Tesseract )
3 Day 7-9 Text parsing and formatting
4 Day 10-13 Chatbot integration with LangChain & Hugging Face
5 Day 14-16 Export functions (CSV, JSON)
6 Day 17-18 User dashboard with document history
7 Day 19-20 Testing, UI polish, mobile responsiveness, deploy

6. **Hosting & Cost Estimation**

Frontend (Vercel) - Free
Backend (Render) - Free tier
OCR Engine (Tesseract) - Open Source

Chatbot (HuggingFace) - Free tier
Authentication & DB (Firebase) - Free tier

7. **Uniqueness & Added Value**

Al-powered document understanding

Conversational search with documents

Free-to-use tech stack for students/startups

Real-world skill development (OCR, LangChain, React)

Mobile optimized and beginner-friendly UI

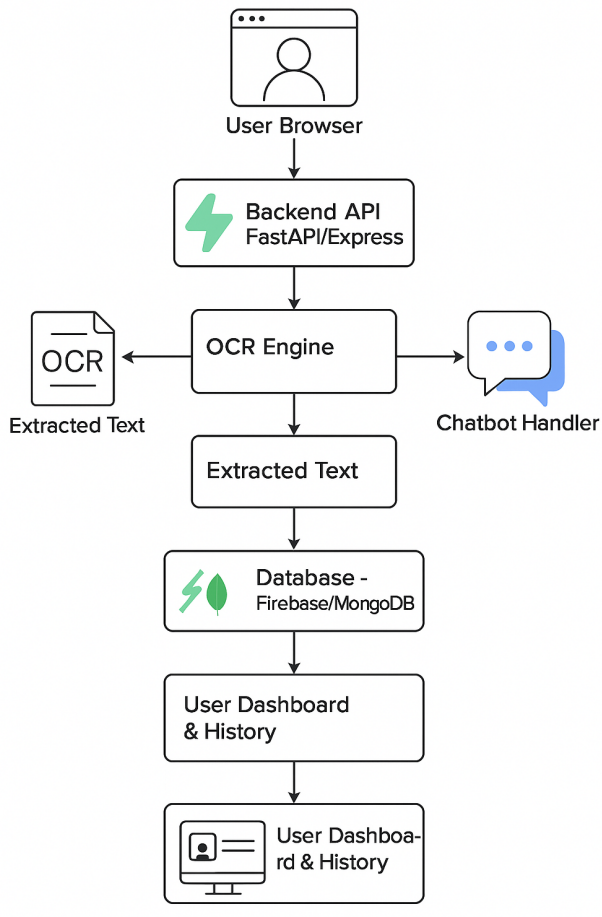
**8. Future Scope**

Support for image-based handwritten documents

Integration with Google Drive / Dropbox

Multi-language OCR and chatbot response

Admin analytics dashboards


User Browser
Backend API
FastAPI/Express
OCR
Extracted Text
OCR Engine
Chatbot Handler
Extracted Text
Database -
Firebase/MongoDB
User Dashboard
& History
User Dashboa-
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